Problem Chosen	2023 ShuWei Cup Summary Sheet	Team Control Number
D		2023102420264

Intelligent Laundry Optimization Under the Synergy of Multiple Models and Algorithms

Abstract : Escalating concerns over global water resources and environmental water source pollution necessitate heightened precision in water resource utilization. Households contribute significantly to water consumption, with washing machines accounting for over one-third of household water usage[1]. This study aims to formulate an optimization model for laundry cleaning[2].

For Problem 1: Initially, we establish a **Monte Carlo simulation solution model**[3] under the criterion that cleanliness is achieved when the remaining amount of dirt is reduced to zero. The model is designed to optimize the objective function and quantify three random variables: a_k (representing the initial amount of dirt), and the amounts of water available. Through meticulous examination, we determine that, when a_k is 0.8, the optimal cleaning method involves a single washing cycle with a residual dirt amount of 0g, achieved with 1.25g of available water. Qualitative analysis of these variables reveals that higher values of a_k, initial dirt, and available water correspond to less dirt remaining after each wash.

For Problem 2: Employing a **dynamic programming algorithm**[4], we address the challenge of formulating the optimal washing scheme to ensure the dirt residue remains below one thousandth of the initial dirt amount. Calculations yield the most time-efficient washing solutions, such as one washing cycle for 2g of initial dirt, consuming 2.5g of water, and one washing cycle for 3g of initial dirt, consuming 3.75g of water. A Monte Carlo simulation model is then employed for qualitative analysis, revealing that larger values of a_k and initial dirt lead to more time-efficient washing solutions, while larger initial dirt amounts result in increased time consumption.

For problem 3: Commencing with **data preprocessing** to obtain the total amount of each dirt substance, we address the issue of detergent usage. For scenarios involving a single detergent, a **mixed integer planning (MIP) model**[5] is constructed and solved using the bisection method[6]. When two detergents are mixed, a **mixed integer nonlinear planning model**[7] is established, and the **branch-and-bound method**[8] is applied to determine the optimal washing solution. The optimal washing solution, when mixing two detergents, involves using the fifth and sixth detergents, with quantities of 24.30g and 31.18g, respectively, resulting in a total cost of 5.35 yuan and a water consumption of 4216.06g.

For Problem 4:

Using a **recursive method**, we identify all washing schemes that satisfy the minimum number of washing cycles. Building upon the models and methods established in Problem 3, we compare the use of a **single detergent and mixed detergents**. The most cost-effective scheme involves a single detergent washing process: the first washing cycle employs Detergent 2 for Silk and Rayon clothing (171.04g), the second cycle for Cotton and Pure Wool clothing (166.50g), the third cycle for Cotton Linen and Cotton Polyester clothing (169.53g), and the fourth cycle for Polyester and Wool Polyester clothing, using Detergent 5 (151.52g). The total water usage is 47,222.36g, with a minimum cost of 24.14 RMB.

Keywords: laundry optimisation, Monte Carlo simulation, dynamic programming, mixed-integer programming models, mixed-integer nonlinear programming models

Content

Abstract 1
1. Introduction 2
1.1 Background 2
1.2 Problem Restatement 2
2. Problem analysis 3
2.1 Analysis of question one 3
2.2 Analysis of question two 4
2.3 Analysis of question three 4
2. 4 Analysis of question four 5
3. Model Assumptions 5
4. Symbol Description 6
5. Modelling and solving 6
5.1 Problem 1 modelling and solution 6
5.1.1 Modelling optimisation in Monte Carlo simulation 6
5.1.2 Model solving 7
5.1.3 a_k , initial fouling, and water availability factors 8
5.2 Problem 2 modelling and solution 10
5.2.1 Modelling dynamic planning 10
5.2.2 Model solving 10
5.2.3 Effect of a_k and initial fouling on the optimal solution 11
5.3 Problem 3 modelling and solution 12
5.3.1 Data pre-processing 12
5.3.2 Single Detergent Selection Models 13
5.3.3 Mixed Detergent Selection Models 14
5.4 Problem 4 modelling and solution 16
5.4.1 Clothing materials and washing programme grouping 16
5.4.2 Modelling 17
5.4.3 Single detergent situation 18
5.4.4 Mixed detergent situation 19
5.4.5 Comparison of detergent mixing methods 21
6. Models Evaluation 21
6.1 Model Evaluation 21
6.2 Model Weakness 22
7. model Promotion 23
8. References 23
Appendix 24

1. Introduction

1.1 Background

The confluence of climate change, global population expansion, and escalating living standards imposes substantial strain on natural resources, particularly jeopardizing the reliable access to water—a vital survival resource. Uninterrupted access to high-quality drinking water stands as a fundamental necessity for daily life, sustenance of food production, industrial processes, and preservation of natural ecosystems[9]. Consequently, there exists an imperative for further research and exploration into methods of laundry cleaning that are not only more time-efficient and economically viable but also align with environmentally sustainable practices, addressing the escalating demands of contemporary living while fulfilling the environmental requisites of sustainable development.

The rapid pace of modern society has catalyzed a heightened focus on the convenience and efficiency of daily life. Laundry cleaning, integral to the fabric of everyday existence, necessitates adaptation to this demand for efficiency. Consequently, the research and development of optimization models for laundry cleaning become paramount, aiming to enhance efficiency, diminish energy consumption, and mitigate environmental impact. In the present societal landscape, the optimization of laundry cleaning processes stands as a critical pursuit.

Simultaneously, burgeoning environmental consciousness prompts an increased concern regarding the environmental implications of laundry cleaning processes. Thus, the exploration of an environmentally friendly laundry cleaning model emerges as an imperative of contemporary society. Through the investigation of innovative detergents, washing methodologies, and energy utilization techniques, we aim to curtail environmental pollution in the laundry cleaning process, thereby contributing to the realization of sustainable development objectives.

1.2 Problem Restatement

This inquiry centers on the development of an optimization model tailored to laundry cleaning, specifically addressing the following challenges:

Question One:

Given the initial amount of dirt and water, alongside the solubility of dirt in water at the kth wash represented as a_k, where $a_1 = 0.80$, $a_k = 0.5a_{k-1}$ for k = 2, 3, ..., we construct a Monte Carlo simulation solution model to optimize the objective function. The objective is to determine the optimal cleaning method under these conditions, quantifying three random variables: a_k, the initial dirt amount, and the available water. The optimal solution is obtained, followed by a qualitative analysis of the influence of a_k, initial dirt amount, and water availability on the objective, utilizing the established model.

Question Two:

Assuming uniform washing times and an abundant water supply, with the final dirt

residue not exceeding one thousandth of the initial dirt volume, and conditions akin to Problem 1, we employ a dynamic programming algorithm. This is to derive the most time-efficient cleaning schedule. Additionally, a Monte Carlo simulation solution model is utilized for qualitative analysis, investigating the impact of a_k and the initial dirt amount on the optimal solution.

Question Three:

Considering various types and quantities of dirt on clothes (Table 1), and with ten available detergents featuring different solubility and unit prices (Table 2), we introduce a cost-effective approach. In scenarios involving no detergent mixing, a dichotomous solving method is applied. For two-by-two detergent mixing, a mixed-integer nonlinear programming and branch-and-bound method are employed. Both approaches yield solutions that are not only cost-saving but also effective in cleaning.

Question Four:

Incorporating diverse materials and dirt types for different clothing items (Table 3), and accounting for restrictions on washing mixtures based on materials and colors (Table 4), we employ a recursive method. This method identifies all washing schemes meeting the minimum number of washes under conditions analogous to Problem 2. Building on the established model and method from Problem 3, we explore the use of single detergents and detergent mixtures. The goal is to ascertain the most economical detergent for each wash, presenting a cost-effective and efficient laundry cleaning plan.

2. Problem analysis

2.1 Analysis of question one

(1) Problem Decomposition:

The initial question is bifurcated into two distinct problems. The first problem necessitates the determination of the optimal cleaning method based on the given dirt and water quantities, excluding other variables. The second problem delves into the analysis of the effects of a_k, initial dirt, and water availability on the objective.

(2) Objective Definition:

The criteria for cleanliness are established, with the residual dirt amount being set to zero. A Monte Carlo simulation solution model is constructed, incorporating a_k, initial dirt, and water availability as random parameters. The objectives are defined as the number of washing cycles and the remaining dirt amount. The model is utilized to derive total water consumption and washing cycle counts for various combinations of initial dirt, washing cycles, and available water. The simulation results are then visually represented.

(3) Monte Carlo Simulation Application:

The Monte Carlo simulation model employs a_k, water availability, and initial dirt as random variables to ascertain the optimal number of washes and the water quantity per wash for each parameter set. Conclusions are drawn qualitatively from the random data, and the simulation results are visualized. This methodology facilitates a comprehensive understanding of the optimal cleaning method under varying conditions, enhancing the

interpretability of the findings.

2.2 Analysis of question two

(1) Problem Decomposition:

Similar to the structure of Problem 1, the second question is also bifurcated into two distinct problems. Building upon the foundation laid in Problem 1, additional conditions are introduced: uniform washing times, unlimited water availability, and a constraint that the final dirt residue must not exceed one-thousandth of the initial dirt amount. The objective is to identify the most time-efficient cleaning method and analyze the impact of a_k and the initial dirt amount on the optimal solution.

(2) Time-Saving Cleaning Method:

In the first sub-question, uniform washing times are assumed, emphasizing the reduction of the total number of washing cycles to optimize time efficiency. Through quantitative integration of the initial dirt amount, specific calculations, visualization, and data processing, the most time-saving cleaning scheme is determined. For instance, when the initial dirt amount is 1g, washing once with 1.25g of water per cycle; when it is 2g, washing once with 2.5g of water per cycle; and when it is 3g, washing once with 3.75g of water per cycle.

(3) Monte Carlo Simulation for Impact Analysis:

The second sub-question employs the Monte Carlo simulation method, setting a_k and the initial dirt amount as random variables. The focus is on analyzing their effects on the number of washes. Through the exploration of random data, conclusions are drawn and subsequently visualized. This approach enables a nuanced understanding of how a_k and the initial dirt amount influence the optimal solution, providing insights into the dynamic relationships between these variables and the washing process.

2.3 Analysis of question three

(1) Optimization Scenarios for Non-Mixed Detergents:

In the first scenario, detergents are not mixed. A model is constructed to optimize the amount of each detergent required for cleaning each type of dirt. The optimization factors include the solubility of the detergent, the initial amount of dirt, and the cost of the water bill.

(2) Data Pre-processing and MIP Model:

Prior to analysis, the data in Table I undergoes pre-processing to obtain the total amount of each dirt substance. Subsequently, a Mixed Integer Programming (MIP) model is formulated. The model is solved using the bifurcation method, ensuring that the cleaned dirt does not surpass one-thousandth of the initial dirt amount. The outcome provides the optimal dosage for each detergent, the total water consumption, and the minimum overall cost. This approach aims to achieve a solution that is both cost-effective and efficient in cleaning.

(3) MILP Modeling for Two-by-Two Detergent Mixing:

In the second scenario, the focus shifts to the use of Mixed Integer Nonlinear Programming (MILP) for modeling and solving. Employing the branch-and-bound method, the objective is to find the optimal mixing solution in the case of two-by-two

combination mixing of detergents. The optimization targets include minimizing the overall cost and reducing water consumption. The final results aim to present a solution that not only achieves cost savings but also minimizes environmental impact through efficient use of water resources.

2. 4 Analysis of question four

(1) Development of Washing Programs for Different Materials:
Considering that certain materials cannot be washed simultaneously, distinct washing programs are devised for each of the eight materials. Following this, an evaluation is conducted to determine the most affordable washing program among all the developed programs.
(2) Recursive Method for Washing Programs:
Initially, data processing of Table IV is undertaken to identify mixed combinations of different clothing materials. Subsequently, a recursive method is employed to identify all washing programs satisfying the minimum number of washes. The total dirt amount for each material is calculated by treating each material as a cohesive entity. Building upon the model and method established in the third question, a solution is derived. The use of both single detergent and detergent mixtures is discussed to present cost-effective and efficient cleaning alternatives.
(3) Comparative Analysis of Detergent Mixing Options:
A comparative analysis is conducted between the two-by-two detergent mixing option and the single detergent option. The objective is to discern and determine the most affordable washing solution. The comparison encompasses considerations of cost, efficiency, and overall effectiveness, providing insights into the optimal approach for achieving economical and efficient laundry cleaning.

3. Model Assumptions

1. Uniform Distribution of Dirt and Dissolution: The assumption is made that dirt is uniformly distributed on clothes, and the washing action leads to complete and uniform dissolution in water [10].
2. Even Distribution of Detergent: It is assumed that the detergent is evenly distributed on the clothes.
3. Detergent Amount Determination: The detergent quantity is based on the laundry weight and degree of soiling, ensuring optimal stain removal effectiveness [11].
4. Ignoring External Factors: The effects of water temperature, water quality, and other external factors on the rinsing process are disregarded.
5. Ignore factors such as the weight and volume of the clothing material.
6. It is assumed that different types of detergents can mix with each other.

4. Symbol Description

Symbol	Description	Unit
a_k	Solubility of dirt in water at the *kth* wash	g
D_k	Amount of dirt remaining after the *kth* wash	g
W_k	Amount of water used per wash	g
n	Number of washings	vice
C_i	Detergent i unit price	Yuan/g
W_p	unit cost of water	Yuan/tonne
σ_{ij}	Solubility of detergent i for dirt substance j	g
x_{ij}	Amount of detergent i used in cleaning of soiled material j	g
W_{total}	Total volume of water used for k washes	g
C_{total}	Total cost of detergent and water for k washes	Yuan

5. Modelling and solving

5.1 Problem 1 modelling and solution

5.1.1 Modelling optimisation in Monte Carlo simulation

(1) Model definition

Monte Carlo algorithm is a numerical computation method based on random sampling, the core idea of which is to simulate the problem by using a large number of random samples in order to estimate unknown values or compute complex mathematical expressions [12].

(2) Establishment of the objective function

Define total water use as the objective function, i.e. to minimise total water use:

$$\text{Minimize} W_{total} = \sum_{k=1}^{n} W_k$$

(3) Setting constraints

$$\begin{cases} n \in Z_+, n \geq 1 \\ D_k = (1 - a_k)D_1 \\ D_k \geq 0 \\ a_k = 0.5a_k - 1 \\ W_k = \dfrac{D_k}{a_k} \\ K = 1,2,3, \dots N \end{cases}$$

(4) Mathematical Modelling

Objective function:

$$\boldsymbol{min_{n,W_k} W_{total} = min_{n,W_k} \sum_{k=1}^{n} W_k}$$

5.1.2 Model solving

(1) Algorithm design

A Monte Carlo simulation algorithm was used to randomly generate values for the initial amount of dirt, the number of cleanings, and the amount of water used to calculate the total water consumption.

(2) Algorithmic ideological basis

The total water consumption under different combinations is obtained by randomly simulating several experiments, from which the optimal solution is selected.

(3) Specific steps

1. Randomly generate the initial amount of dirt D_1 , the number of washings n, the solubility a_k , and calculate the amount of water used each time W_k and the total amount of water used W_{total} .

2. Repeat the above steps several times to record the parameter combination corresponding to the smallest total water consumption.

3. Monte Carlo simulations yield some of the following data (see appendix tables 1-4 for complete data charts).

Initial Dirt	Max Washes	Optimal Washes	Optimal Water Amounts	Total Water Amount
1	1	1	[1.25]	1.25
2	9	1	[2.5]	2.5
3	7	1	[3.75]	3.75
4	6	1	[5.0]	5
5	2	1	[6.25]	6.25

Table 1-1 Optimal washing methods for simulating different initial dirt quantities and number of washes

4. Visualization of data

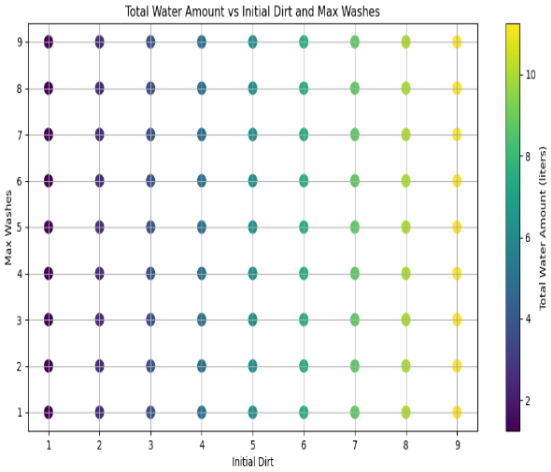

Figure 1-2 Total Water Volume

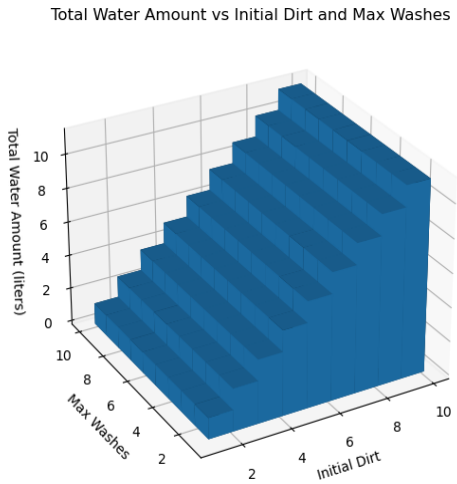

Figure 1-3 Total water consumption

(4) Conclusion

From the simulation results, it can be obtained When the initial amount of fouling is 1g, the optimal cleaning method is the number of cleaning times is 1 and the amount of water used per time is 1.25g , i.e., it is shown in Fig. 1-4.

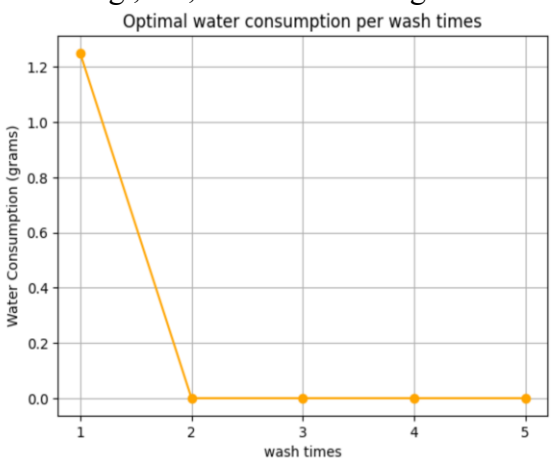

Figure 1-4 Optimum water consumption per wash

5.1.3 a_k , initial fouling, and water availability factors

We again discuss the effect of a_k, initial fouling and water availability on the target by Monte Carlo simulation methods with randomly taken values for solubility a_k , initial fouling and water availability.

(1) Modelling steps

1. Monte Carlo simulation

For the given parameters, multiple Monte Carlo simulations were performed. Initial conditions were randomly generated to simulate the washing process and the water usage per wash was recorded. The optimal number of washes and water consumption per wash for each set of parameter values is found by comparing the total water consumption.

2. visualization

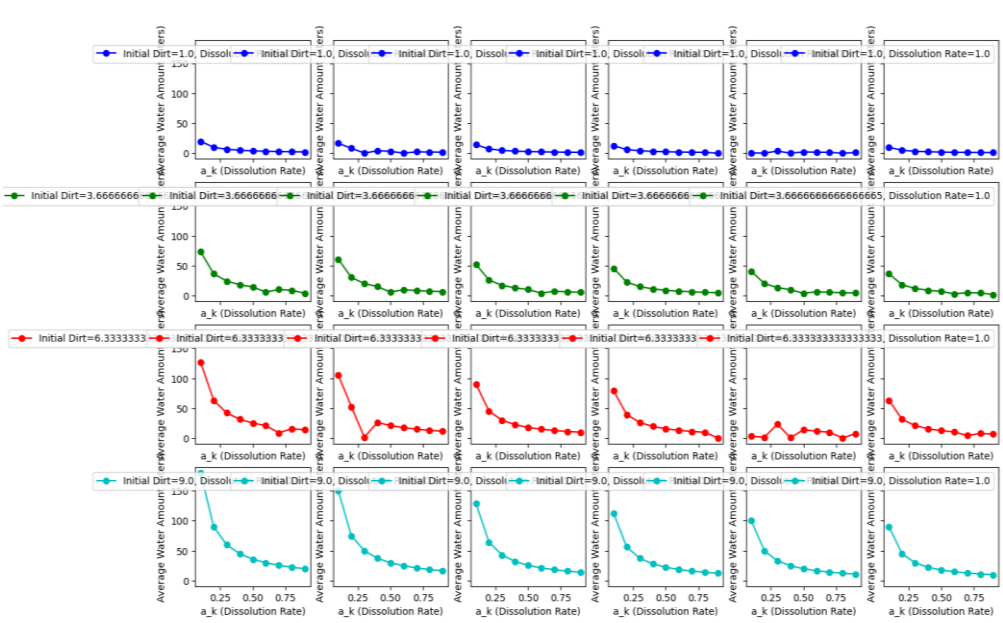

Figure 1-5 Target change under the influence of the triple random variable

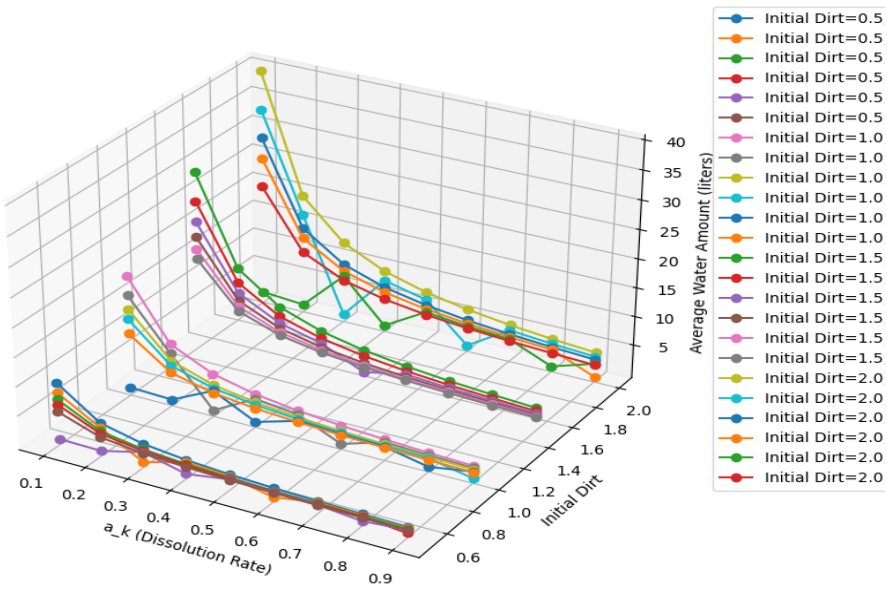

Figure 1-5 Effect of dissolution rate and initial fouling on water consumption

Different combinations of parameters were traversed through nested loops. For each set of parameters, Monte Carlo simulations were performed and the average water use per use was recorded to show the effect of different parameter values on the average water use per use. Each subplot shows the average water use per use for different values of a_k for a specific set of initial fouling and water availability.

(2) Conclusion

The effect of a_k , initial fouling and water availability on the average water consumption per use is visualised by means of 3d graphs. The trend of water consumption at different parameter values can be observed.

It can therefore be concluded that, other things being equal, the greater the a_k , the less dirt remains after each wash; the greater the initial amount of dirt, the less dirt

remains after each wash; and the greater the amount of water available, the less dirt remains after each wash.

5.2 Problem 2 modelling and solution

5.2.1 Modelling dynamic planning

（1）Model Definition

Dynamic Programming (DP) is an algorithmic idea that solves problems by splitting them into overlapping subproblems and solving each subproblem only once. The basic idea is to reduce the computational complexity by decomposing the original problem into a number of subproblems and solving each subproblem only once and saving the solution to avoid repeated computations [4].

(2) Setting constraints

The basic model is similar to Problem 1, but with the addition of constraints on the amount of dirt residue.

$$\begin{cases} n \in Z_+, n \geq 1 \\ D_k = (1 - a_k)D_1 \\ D_k \geq 0 \\ a_k = 0.5a_{k-1} \\ W_k = \dfrac{D_k}{a_k} \\ K = 1,2,3, \dots N \\ D_n \leq 0.001D_1 \end{cases}$$

(3) Mathematical modelling

Objective function:

$$\boldsymbol{min_{n,W_k} W_{total} = min_{n,W_k} \sum_{k=1}^{n} W_k}$$

(4) Simplified Modelling

The problem is solved using the dynamic programming method by splitting the problem into sub-problems: choosing the right amount of water to be used during each wash so that the total water consumption is minimised.

5.2.2 Model solving

(1) Algorithm design

Dynamic programming (DP) algorithms are used to solve the optimal cleaning solution, ensuring that the amount of dirt remaining is no more than one thousandth of the initial amount of dirt.

(2) Algorithmic ideological basis

Dynamic programming algorithms are suitable for problems with optimal substructures, where the problem is divided into sub-problems and the optimal solutions of the sub-problems are recorded to construct an optimal solution to the overall problem.

(3) Specific steps

1. Establish a state transfer equation to calculate the amount of dirt remaining after each

wash.

2. Use dynamic planning algorithms to progressively calculate the optimal cleaning solution to ensure that the amount of dirt residue does not exceed the specified threshold.

3. By dynamic planning can be obtained as Table 2-1 part of the data (complete data charts see appendix Table 2-5).

Initial Dirt	Dissolution Rate	Optimal Water Amount
1	0.8	1.25
2	0.1	20
3	0.7	4.286
4	0.2	20
5	0.5	10

Table 2-1 Optimum water consumption for different fouling amounts and solubilities

(4) Conclusion

The most time-saving cleaning programme is: when the initial amount of dirt is 1g, 1 wash with 1.25g of water per wash; when the initial amount of dirt is 2g, 1 wash with 2.5g of water per wash; and when the initial amount of dirt is 3g, 1 wash with 3.75g of water per...

Relationship between initial fouling amount, solubility, and water consumption under the optimal solution:

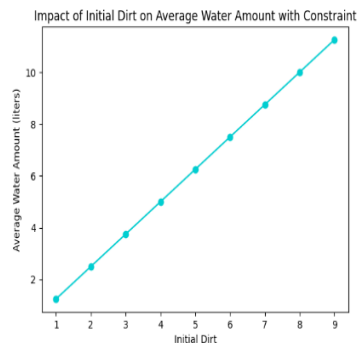

Figure 2-1 Initial amount of fouling vs. amount of water used per time

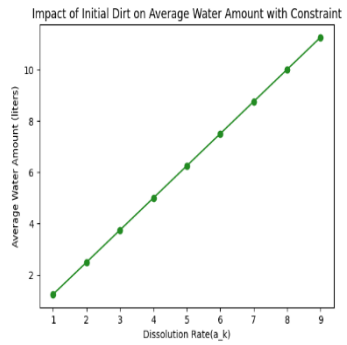

Figure 2-2 Relationship of ak per water use

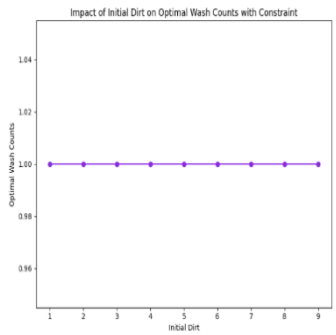

Figure 2-3 Impact of Initial Dirt on Optimal Wash Counts with Constraint

5.2.3 Effect of a_k and initial fouling on the optimal solution

(1) Data visualization

Data visualization using Monte Carlo simulation.

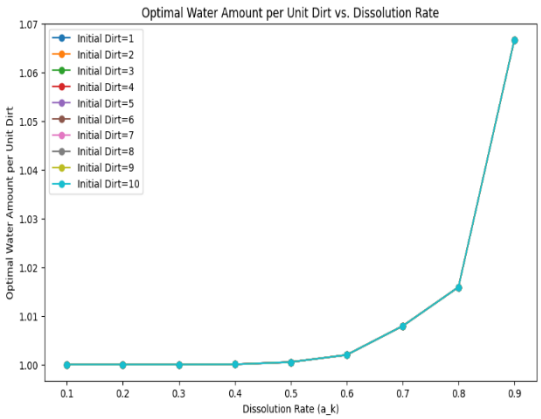

Figure 2-4 Optimum amount of water

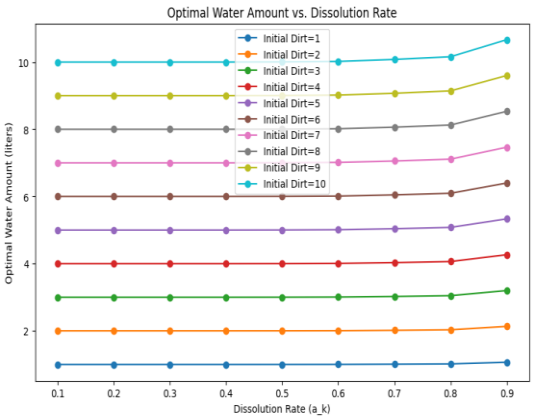

Figure 2-5 Optimum water consumption

（2）Analysing charts

obtained from image analysis:

1. Effect of dirt *solubility* a_k

Analysis: The results show that for a given initial amount of dirt, the dirt solubility a_k has an effect on the average amount of water used per wash for the optimal solution. A higher a_k indicates that the dirt dissolves more easily, resulting in less water being used for cleaning. Therefore, in practical applications, the selection of a cleaning agent or method with a higher solubility may help to reduce the amount of water used.

2. Effect of initial fouling volume

Analysis: The results show that an increase in the initial amount of dirt leads to an increase in the average amount of water used per pass. This is reasonable because more fouling requires more water volume for cleaning. Therefore, reducing the amount of initial fouling may be a strategy to reduce water usage in practical applications.

（3）reach a verdict

In short, as shown in Figure 2-6: when other conditions are certain, the larger a_k is, the fewer the total number of cleaning times, the less the total cleaning time, the more time-saving; the smaller the initial amount of dirt, the fewer the total number of cleaning times, the less the total cleaning time, the more time-saving.

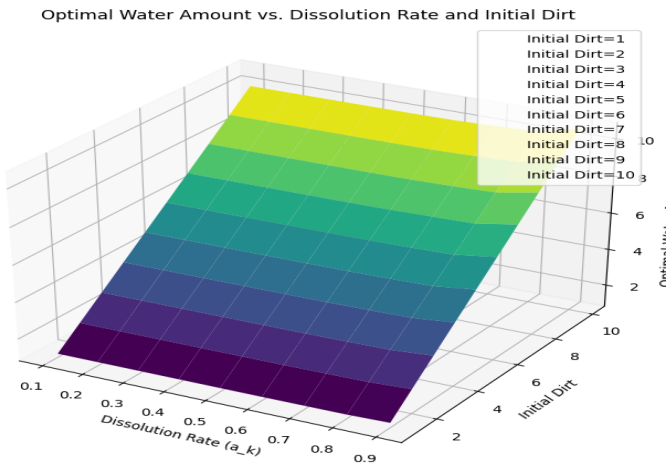

Figure 2-6 Relationship between optimum water volume and solubility and initial fouling

Overall, fouling solubility and initial fouling volume play an important role in the optimal cleaning programme. Selecting a detergent with high dirt solubility and controlling the initial dirt volume can help to reduce the amount of water used per pass, thereby increasing the efficiency of the cleaning process. In practice, these factors can be adjusted on a case-by-case basis to balance cleaning effectiveness and resource utilisation.

5.3 Problem 3 modelling and solution

5.3.1 Data pre-processing

Firstly, data preprocessing was performed on Table 1 given in the title to obtain the following Table 3-1.

Garment	Contaminant 1	Contaminant 2	Contaminant 3	Contaminant 4	Contaminant 5	Contaminant 6	Contaminant 7	Contaminant 8
	116	91	96	106	99	71	86	76

Table 3-1 Type and amount of overall soiling of the clothes

5.3.2 Single Detergent Selection Models - Based on Mixed Integer Programming

(1) Establishment of a mixed integer planning MIP model

1. Modelling assumptions

1. Residue solubility assumption: It is assumed that the solubility relationship s between the detergent and each type of residue during a single wash is linear.

2. Assumption of washing standard: It is assumed that the washing standard of each residue is one thousandth of its initial content, i.e. the residue of the residue should not exceed one thousandth of the initial content.

2. Constraints

$$\begin{cases} n \in Z_+, n \geq 1 \\ D_k = (1 - a_k)D_1 \\ D_k \geq 0 \\ W_k = \dfrac{D_k}{a_k} \\ K = 1,2,3, \dots N \\ D_n \leq 0.001D_1 \\ W_{total} = W_1 + W_2 + \cdots W_k \\ C_{total} = \sum_{k=1}^{n} W_k W_P + \sum_{j=1}^{8} X_{i,j} C_i \end{cases}$$

(2) Model solving

1. Dichotomous solution

We obtain Table 3-2 based on the dichotomous solution:

	Detergent	Detergent Amount(g)	Cost (yuan)	Water Usage (g)
1	Detergent6	98.602597	9.860549	7493.797387
2	Detergent3	94.90500	10.003077	6456.357679
3	Detergent5	126.540000	10.101346	11489.887477
4	Detergent2	115.036363	10.353562	8742.763619
5	Detergent8	115.036363	11.474079	7405.822044
6	Detergent7	118.631250	14.236039	9015.974982
7	Detergent9	124.465574	22.404092	9459.383587
8	Detergent4	138.043636	22.461120	8393.264983
9	Detergent10	180.771428	24.691679	10213.233313
10	Detergent1	104.005479	26.001659	7904.416422

Table 3-2 Data on the use of single detergents

2. Data visualisation

The data obtained from the solution was visualised to obtain Figure 3-1 Cost of washing clean for each detergent and Figure 3-2 Amount of water required for washing

for each detergent.

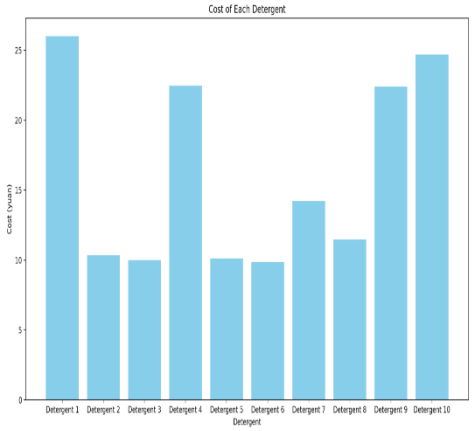

Figure 3-1 Cost of a clean wash for each type of detergent

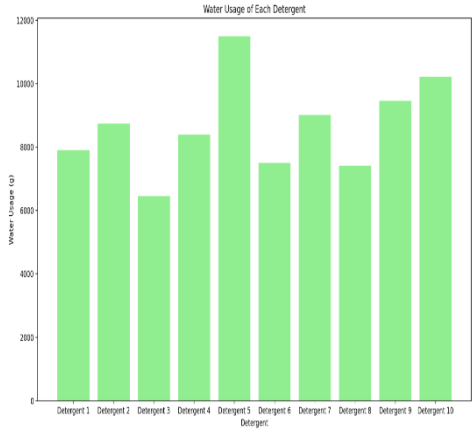

Figure 3-2 Water Required for Each Detergent Wash

3. Conclusion

Based on the tabular information visualization data, it is possible to visualise the cost of using each detergent and the amount of water used to use each detergent. Therefore, the best cleaning solution was arrived at: the 6th detergent was chosen for one wash, at which time the cost used was ￥9.86 and the total water consumption was 7493.80g.

5.3.3 Mixed Detergent Selection Models - Based on Mixed Integer Nonlinear Programming

(1) Model definition

The integer nonlinear programming MINLP (Mixed-Integer Nonlinear Programming) model is a complex optimisation model that introduces integer variables to linear programming (LP) and nonlinear programming (NLP). This type of model is suitable for problems that contain discrete decision variables and nonlinear constraints or objective functions. Overall, the MILP model is more suitable for integrating multiple complex elements of this problem.

(2) Modelling of integer non-linear programming MINLP

1. Decision-making variables

D_k : Amount of dirt remaining after the kth wash.

W_k : Amount of water used per wash.

x_j : whether or not detergent j is selected, xj=1 means selected, x_j =0 means not selected.

2. Constraints

$$\begin{cases} \boldsymbol{D_{k+1} = (1 - a_k) D_k} \\ \boldsymbol{W_k = \frac{D_k}{a_k}} \\ \boldsymbol{W_k \in Z_+} \\ \boldsymbol{X_j \in \{0, 1\}} \\ \boldsymbol{\sum_{j=1}^{10} X_{j \geq 1}} \end{cases}$$

3. Objective function

$$\textbf{Minimize Cost} = \sum_{j=1}^{10}(C_i X_j) + \sum_{k=1}^{10} W_p W_k$$

(3) Model solving

1、Branch delimitation method of solving

For the model we have built, the branch-and-bound method is used to solve the problem, and a flowchart is drawn to make the idea clearer.

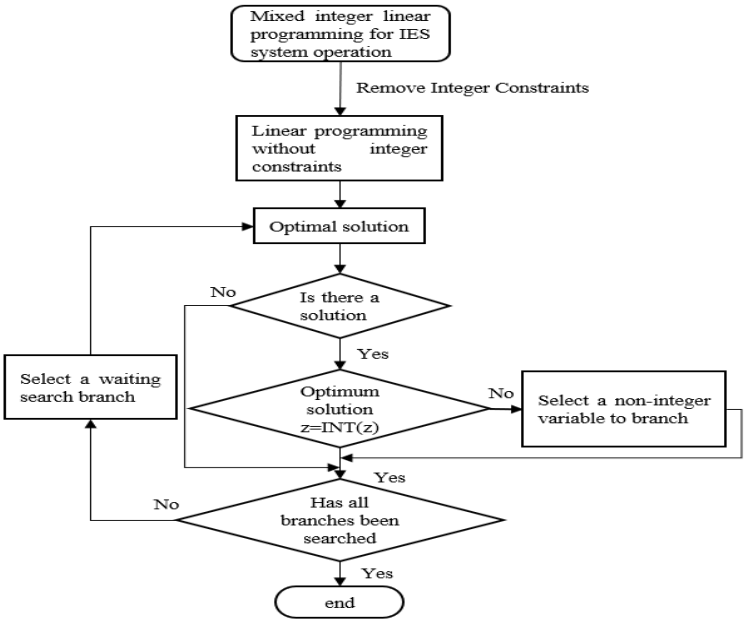

Figure 3-3 Flowchart of branch delimitation method

2、Detergent used in combination of two

See Data Table 3-3 in the following section for the cost and volume of water used, and Data 3-6 in the Annex for complete data.

	Combinations	detergent1 dosage	detergent2 dosage	Total dosage	cost	water_usage
1	[5,6]	24.295380	31.179071	55.474452	5.350338	4216.058331
2	[2,6]	24.529316	28.617536	53.146852	5.358192	4039.160779
3	[3,6]	24.666315	23.741328	48.407643	5.376227	3678.980837
4	[2,5]	31.594860	28.722600	60.317459	5.430145	4584.126916
5	[2,3]	23.531619	28.523175	52.054794	5.544195	3956.164325

Table 3-3 Costs and volume of water used for two-by-two combinations of detergents

3. Data visualization

Based on the tabular information visualization data, it is possible to visualise the cost of using each mixed detergent combination and the amount of water used to use each detergent.

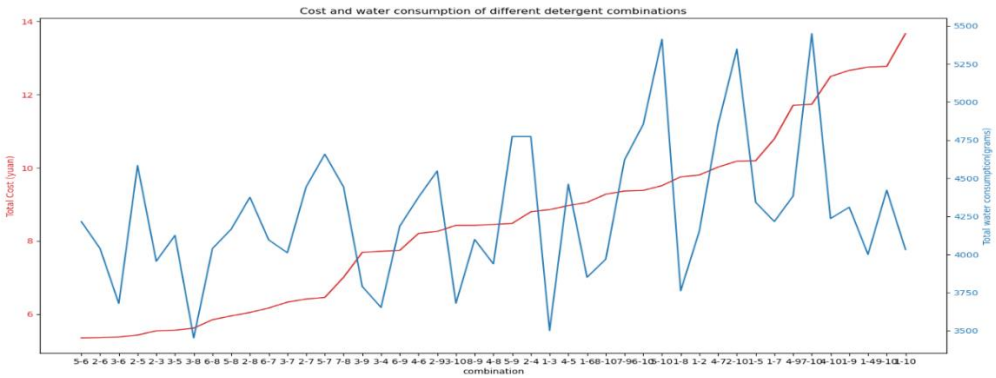

Figure 3-4 Cost Volume and Water Use for Different Detergent Combinations

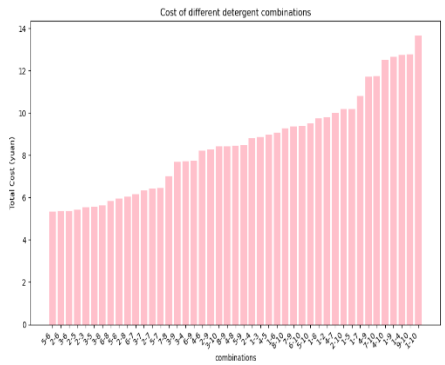

Figure 3-5 Cost Volume of Different Detergent Combinations

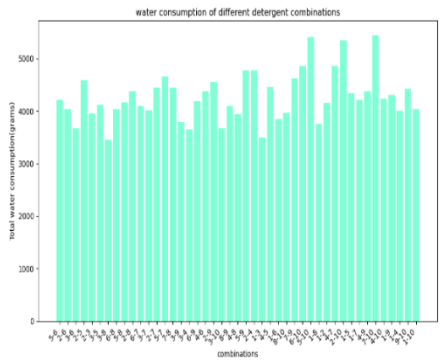

Figure 3-6 Water consumption for different detergent combinations

4. Conclusion

According to the graphical analysis, the optimal solution that saves cost and energy can be derived: Choosing a mixture of the 5th and 6th detergents, the 5th detergent used is 24.30g and the 6th detergent used is 31.18g, for one wash, the total cost used is 5.35 RMB and the total water consumption is 4216.06g.

5.4 Problem 4 modelling and solution

5.4.1 Clothing materials and washing programme grouping

Table 3 given in the title was subjected to data processing to obtain Table 4-3 Types of pollutants and corresponding quantities on clothes of each material.

Material	Contaminant 1	Contaminant 2	Contaminant 3	Contaminant 4	Contaminant 5	Contaminant 6	Contaminant 7	Contaminant 8
1	39	39	52	60	64	44	58	54
2	86	77	57	60	83	66	67	64
3	79	65	57	63	44	55	47	70
4	62	62	62	63	69	80	77	46
5	45	69	29	44	45	53	68	56
6	53	45	28	28	37	41	29	42
7	45	38	40	55	40	53	39	45
8	58	66	37	43	55	52	46	49

Table 4-3 Types of pollutants and corresponding quantities on clothes of each material

Based on Table 4 given in the title, data preprocessing was carried out to obtain Table 4-1 Mixing between clothes of different materials .

Materia	1	2	3	4	5	6	7	8
1	1	0	1	1	1	1	0	0
2	0	1	0	0	0	0	0	1
3	1	0	1	0	0	1	1	1
4	1	0	0	1	0	1	1	1
5	1	0	0	0	1	0	1	1
6	1	0	1	1	0	1	1	1
7	0	0	1	1	1	1	1	1
8	0	1	1	1	1	1	1	1
Note: 1. Material 1 - Cotton, Material 2 - Silk, Material 3 - Cotton Linen, Material 4 - Polyester, Material 5 - Pure Wool, Material 6 - Cotton Polyester, Material 7 - Wool Polyester, Material 8 - Rayon Material 7 - Wool Polyester, Material 8 - Rayon								

Table 4-1 Mixing between clothes of different materials (1 is mixable, 0 is not mixable)

Recursive method for finding the optimal grouping scheme

Based on the modelling of the first three questions, it can be seen that the number of washes has a greater impact on the economic efficiency than the amount of water used, i.e. the most affordable washing solution has the least number of total washes. Using the recursive method, the minimum number of washes can be obtained as 4. There are four scenarios as shown in Table 4-2 .

	Grouping of materials for the 1st wash	Grouping of materials for the 2nd wash	Grouping of materials for the 3rd wash	Grouping of materials for the 4th wash
Option 1	[2,8]	[1,3]	[4,6]	[5,7]
Option 2	[2,8]	[1,4]	[3,6]	[5,7]
Option 3	[2,8]	[1,5]	[3,7]	[4,6]
Option 4	[2,8]	[1,5]	[3,6]	[4,7]

Table 4-2 Grouping of materials for mixed washable garments

5.4.2 Modelling

Based on the modelling and methodology developed in Question 3, the use of single detergents and mixed detergents are discussed separately, assuming that different types of detergents can be mixed with each other. Unlike the model developed in Question 3, this question also needs to consider the effect of clothing material. Therefore, it is necessary to group clothes of different materials.

Set the amount of the kth detergent in each combination at the jth wash to be X_{kj} (k=1,2,...,10), the total amount of water used in each combination W_i , and the total cost of each combination C_l .

Minimise the total cost of each combination by considering the detergent selection and dosage for each combination, and the corresponding cost and total water usage.

5.4.3 Single detergent situation

(1) Discuss each of the four washing programmes

1. Programme I

In a single detergent selection model, optimizing the washing cycles resulted in cost-effective outcomes. The most economical detergents for each wash were determined: detergent 2 for the first and fourth washes, detergent 5 for the second wash, and detergent 6 for the third wash. The cumulative water consumption for the four washes amounted to 50,524.54g, with a total cost of ¥47.8.

2. Programme II

In the single detergent selection model, the optimized washing cycles yielded cost-efficient results. Detergent 2 was the most economical choice for the first, third, and fourth washes, while detergent 5 proved optimal for the second wash. The cumulative water consumption for these four washes was 48,855.74g, with a total cost of ¥36.05.

3. Programme III

In the single detergent selection model, optimizing the washing cycles resulted in cost-effective outcomes. Detergent 2 was identified as the most economical choice for the first three washes, while detergent 6 was optimal for the fourth wash. The cumulative water consumption for these four washes was 34,936.47g, with a total cost of ¥46.23.

4. Programme IV

In the single detergent selection model, the optimized washing cycles resulted in cost-effective outcomes. Detergent 2 was identified as the most economical choice for the first three washes, while detergent 5 was optimal for the fourth wash. The cumulative water consumption for these four washes was 47,222.36g, with a total cost of ¥24.14.

(2) Data visualization

Visualising the data for the four scenarios yields Figure 4-1.

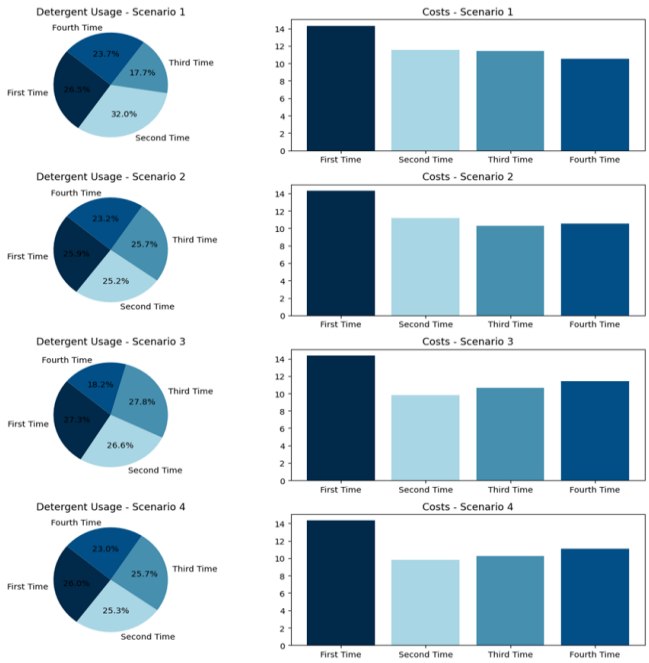

Figure 4-1 Percentage of Detergent Used and Spent per Wash

(3) Conclusion

The laundering regimen encompasses four distinct cycles:

Initial laundering involves Silk clothing and Rayon clothing utilizing detergent 2, with a detergent dosage of 171.04g.

Subsequent laundering entails Cotton clothing and Pure Wool clothing employing detergent 2, with a detergent dosage of 166.50g.

A third laundering iteration encompasses Cotton Linen clothing and Cotton Polyester clothing utilizing detergent 2, featuring a detergent dosage of 169.53g.

The fourth laundering episode involves Polyester clothing and Wool Polyester clothing employing detergent 5, with a detergent dosage of 151.52g.

This optimized laundering plan results in a cumulative water consumption of 47,222.36g, achieving the minimal cost of ¥24.14.

5.4.4 Mixed detergent situation

(1) Discuss each of the four options

1. Programme I

The optimal detergent combination selected for the 1st wash was (5,6), with 36.12 g of detergent 5 and 46.36 g of detergent 6, costing ￥7.96 and using 9320.44 g of water.

The optimal detergent combination selected for the 2nd wash was (2,5), with 36.06 g of detergent 2 and 38.76 g of detergent 5, costing ￥6.74 and using 7781.29 g of water.

The optimal detergent combination selected for the 3rd wash was (5,6) with 28.13 g of detergent 5 and (36.10) g of detergent 6, costing ￥6.20 and using 5652.55 g of water.

The optimal detergent combination selected for the 4th wash was (2,8), with 27.35 g of detergent 2 and 29.21 g of detergent 8, costing ￥5.94 and using 3902.46 g of water.

Therefore, the total water consumption for the first option is 26,656.74g and the total cost is ￥26.84.

2. Programme II

The optimal detergent combination selected for the 1st wash was (5,6), with 36.12g of detergent 5 and 46.36g of detergent 6, costing ￥7.96 and using 9320.44g of water.

The optimal detergent combination selected for the 2nd wash was (2,5), with 35.02g of detergent 2 and 37.64g of detergent 5, costing ￥6.55 and using 7338.85g of water.

The optimal detergent combination selected for the 3rd wash was (2,5), with 42.01g of detergent 2 and 27.22g of detergent 5, costing ￥6.27 and using 5607.69g of water.

The optimal detergent combination selected for the 4th wash was (2,8), with 27.35g of detergent 2 and 29.21g of detergent 8, costing ￥5.94 and using 3902.46g of water.

Therefore, the total water consumption for the second option is 26,169.44g and the total cost is ￥26.72.

3. Programme III

The optimal detergent combination selected for the 1st wash was (5,6), with 36.12g of detergent 5 and 46.36g of detergent 6, costing ￥7.96 and using 9320.44g of water.

The optimal detergent combination chosen for the 2nd wash was (2,3), with 29.93g of detergent 2 and 27.21g of detergent 3, at a cost of ￥6.01 and a water consumption of

4800g.
The optimal detergent combination chosen for the 3rd wash was (2,5), with 43.57g of detergent 2 and 28.23g of detergent 3, at a cost of ￥6.50 and a water consumption of 6030.77g.
The optimal detergent combination selected for the 4th wash was (5,6), with 28.13g of detergent 5 and 36.10g of detergent 6, costing ￥6.20 and using 5652.55g of water.
Therefore, the third option uses a total of 25,803.76g of water and costs a total of ￥26.67.

4. Programme IV
The optimal detergent combination selected for the 1st wash was (5,6), with 36.12g of detergent 5 and 46.36g of detergent 6, costing ￥7.96 and using 9320.44g of water.
The optimal detergent combination chosen for the 2nd wash was (2,3), with 29.93g of detergent 2 and 27.21g of detergent 3, at a cost of ￥6.01 and a water consumption of 4800g.
The optimal detergent combination selected for the 3rd wash was (2,5), with 42.01g of detergent 2 and 27.22g of detergent 5, costing ￥6.27 and using 5607.69g of water.
The optimal detergent combination selected for the 4th wash was (5,6), with 29.09g of detergent 5 and 37.33g of detergent 6, costing ￥6.41 and using 6044.54g of water.
Therefore, the fourth option has a total water consumption of 25772.67g and a total cost of ￥26.65.

(2) Data Visualization

Visualization of the data for the four scenarios.

Figure 4-2 Cost per wash for four scenarios

(3) Conclusion

In summary, the most affordable washing solution when using mixed detergents

is:

The first washing of Silk clothing and Rayon clothing was done using a mixture of detergents 5 and 6 with 36.12g of detergent 5 and 46.36g of detergent 6;

Cotton clothing and Pure Wool clothing were washed a second time using a mixture of detergents 2 and 3, with 29.93 g of detergent 2 and 27.21 g of detergent 3;

The third washing of the materials Cotton Linen clothing and Cotton Polyester clothing was carried out using a mixture of detergents 2 and 5, with a dosage of 42.01 g for detergent 2 and 27.22 g for detergent 5;

The fourth washing of Polyester clothing and Wool Polyester clothing was carried out using a mixture of detergents 5 and 6, with 29.09 g of detergent 5 and 37.33 g of detergent 6.

The total water consumption was 25,772.67g and the total cost was ￥26.65.

5.4.5 Comparison of detergent mixing methods

Comparing the two-by-two detergent mixing scenario here with the single detergent choice scenario shows that the single detergent choice scenario is more affordable than the mixed detergent scenario, so there is no need to discuss the case of a wider variety of detergent mixes.

The data for the optimal washing solution (single detergent) is visualised as follows in Figure 4-3 Graph of the cost per wash for the most affordable washing solution, and in Figure 4-4 Graph of the type and amount of detergent used in the most affordable washing solution.

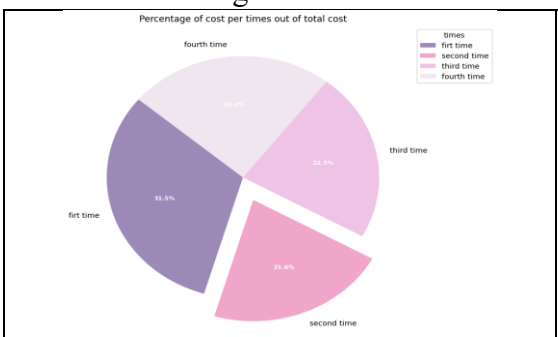

Figure 4-3 Cost per wash graph for the most affordable washing option

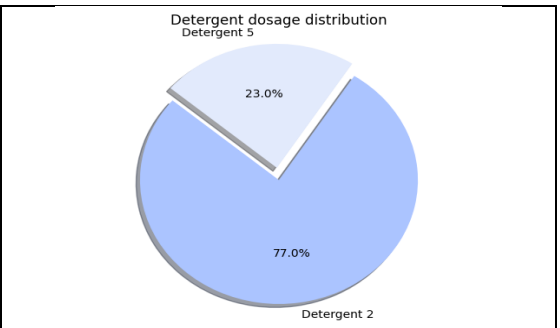

Figure 4-4 Chart of detergent types and amounts used in the most affordable washing programme

6. Models Evaluation

6.1 Model Strengths

1.Monte Carlo Simulation Optimization Models:

- Robust Separation of Physical Effects: These models facilitate the individualized examination of various physical effects, contributing to a nuanced understanding of key factors and their underlying physical nature [13].
- Simulation of Extreme Conditions: Computational simulations enable the emulation of extreme experimental conditions, offering the ability to calculate physical quantities across diverse environments.

- Implementation Simplicity: Exhibiting simplicity in design and execution, these models are particularly well-suited for addressing high-dimensional complex problems.
- Sample Size Enhancement: Precision in results is attainable by augmenting the sample size, thereby bolstering the accuracy of outcomes.

2.Dynamic Programming:

- Effective Subproblem Resolution: Dynamic programming demonstrates efficacy in resolving overlapping subproblems within the scope of the given problem.
- Time Complexity Reduction: The application of dynamic programming leads to a notable reduction in time complexity.
- Optimized Results: The outcomes generated through dynamic programming exhibit optimization, enhancing the overall efficiency of the model.

3.Integer Non-Linear Planning Model:

- Enhanced Flexibility: This model offers increased flexibility in addressing challenges associated with the non-linear correlation between solubility and detergent dosage, including potential non-linear increases in solubility with escalating detergent dosage.
- Simultaneous Optimization of Multiple Objectives: The model allows for the concurrent optimization of multiple objectives, contributing to a comprehensive approach in problem-solving.
- Adaptable Constraint Handling: It demonstrates adaptability in managing various constraints, encompassing the discrete nature of wash counts and non-linear constraints related to detergent solubility.

6.2 Model Weakness

1.Monte Carlo Simulation Optimization Model:

- Convergence Speed Variation: The model exhibits slower convergence for specific problems, potentially impeding efficiency in certain contexts.
- Computational Intensity: A considerable number of random samples are necessitated, rendering the model computationally expensive in terms of resources.
- Applicability Challenges: The model may not be universally applicable, requiring specialized optimization techniques for certain problem domains.

2.Dynamic Programming:

- Storage Space Requirement: The implementation of dynamic programming demands additional storage space, which may pose constraints, particularly in resource-limited environments.
- Limited Applicability: Inapplicability to certain problem types, particularly those lacking optimal substructures, represents a limitation in the scope of dynamic programming.
- Oversimplification of Clothing Characteristics: The model overlooks clothing weight and volume for computational convenience, leading to somewhat coarse conclusions in scenarios where such characteristics are significant.

7. Model promotion

The model developed holds significant applicability in real-world production and daily life. One notable application is the design of water-saving washing machines, aligning with the principle of water conservation[14]. Recommendations are made for washing machine manufacturers to incorporate key findings in the instruction manuals accompanying their products. This includes indicating optimal rinsing frequencies, recommended detergent types and dosages, and corresponding water consumption for different clothing materials. Such guidance, tailored to the regional variations in water costs and detergent availability, serves as a valuable reference for users, optimizing their laundry practices.

8. References

[1] ZHAO Zhiqiang,XU Sheng,LU Peishi et al. Application and industrialisation of flocculation water-saving technology on washing machines[C]//China Household Electrical Appliances Association. Qingdao Haier Washing Machine Co Ltd;State Key Laboratory of Digital Home Appliances;,2020.

[2] Lu Peimeng,Pan Jingjing. Laundry water saving model[J]. Science and Technology Information, 2012,(02).

[3] H.M. Zhou,X.B. Luo,C.L. Ye. A proposed Monte Carlo method for a class of stochastic optimal control problems[J]. Mathematical Theory and Applications,2022,42(03).

[4] YANG Aimin,GUO Yaopeng,WANG Yulian. A dynamic planning model for water-saving laundry[C]//Chinese Research Society of Preferential Method of Integration and Economic Mathematics, Institute of Science and Technology Policy and Management Science, Chinese Academy of Sciences. Department of Mathematics, Shanxi University; Department of Mathematics, Shanxi University; Computing Centre, Shanxi University;, 1999.

[5] Yang Xiaobin. Mixed integer programming properties and its construction of superadditive functions[J]. Journal of Huaqiao University (Natural Science Edition),2016,37(02):257-260.

[6] Li, Dongyu. Dichotomous method for solving the miniaturised maximal problem with constraints[D]. Huazhong University of Science and Technology,2022.

[7] Cristina M M ,Armando C ,Claudia D , et al.A new mixed integer non-linear programming model for optimal PAT and PRV location in water distributionnetworks [J].Urban Water Journal,2021,18(6).

[8] Lin Dang. Research and implementation of efficient algorithm for solving real roots of nonlinear systems of equations based on branch-and-bound method[D]. East China Normal University,2019.

[9] Yongqiang Z ,Congcong L ,S H F C , et al. Southern Hemisphere dominates recent decline in global water availability.[J].Science (New York, N.Y.),2023,.